Nikhil Kotikalapudi

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Education

The Pennsylvania State University – Schreyer Honors College — School of EECS State College, PA
Bachelor of Science in Computer Engineering, GPA: 3.94/4.0 *Expected Graduation: May 2027*

Experience

AMD Austin, TX
Incoming GPU Performance Intern (AI Models) *May 2026*
- Joining the machine learning frameworks team to optimize ROCm GPU kernels for AI model inference performance.
- Designing an automated GPU profiling pipeline overcoming current profiling-level limitations using vLLM and kineto traces.

AlphaGPU Palo Alto, CA
Software Engineer Intern (Infrastructure) *Aug 2025 – Jan 2026*
- Deployed robust GPU profiling for 1000+ users with NVML and torch profiler, overcoming cloud-hosted GPU limitations of NSYS/NCU.
- Scaled LeetGPU, a platform democratizing hardware-accelerated programming via GPU challenges, online test environments, and CLI tools supporting CUDA, Triton, PyTorch, Mojo, and CuTe DSL.

NeuroAI Lab State College, PA
AI Acceleration Research Assistant *Feb 2025 – Present*
- Designed a novel scalable high-dimensional Bayesian optimization algorithm with dynamic sampling for ML hyperparameter tuning.
- Research hardware-software co-design for neuromorphic neural networks and novel accelerated architectures (e.g. ferromagnetic).

IEEE Electronics Packaging Society Piscataway, NJ
Electronics Research Intern *Feb 2025 – May 2025*
- Simulated neuromorphic hardware limitations (R_{ON}/R_{OFF}, bit discretization), sustaining 96.9% model accuracy under constraints.
- Implemented hardware-in-the-loop training with varying weight quantization, limiting accuracy drop to 8% on image data.

nth Solutions Coatesville, PA
Electrical Engineering & Data Analyst Intern *June 2023 – July 2024*
- Developed 3 ML-driven signal processing algorithms and benchmarked 100+ nonstationary time-series in Python and MATLAB.
- Co-inventor on 2 products under patent preparation involving sensor-based safety and vibration reduction systems.

Projects

FlashAttention CUDA Kernel | *CUDA, C++, PyTorch*
- Developed a FlashAttention-inspired attention kernel in CUDA achieving 1.54x speedup and handling 8x more tokens than naive self-attention; scales to 200k+ context on a Tesla T4, benchmarked against PyTorch SDPA.
- Achieved O(N) memory via tiling and online softmax, minimizing HBM accesses; resolved shared-memory bank conflicts with strided padding and tuned FP16 tiling to maximize occupancy and throughput.

CUDA-Accelerated Neural Network from Scratch | *CUDA, C++, Python, NumPy*
- Built a deep neural network from scratch (no ML libraries) reaching 97.83% on MNIST and 88.02% on Fashion-MNIST with a custom optimizer for dynamic learning-rate scheduling.
- Accelerated training 10.57x (40 to 3.8 minutes) via GPU parallelization with a custom CUDA matmul kernel.

Essential Machine Learning Projects | *Python, PyTorch, Transformers, Scikit-Learn*
- Built a RAG bot with the TinyLlama LLM, a pneumonia-detection CNN reaching 90%+ accuracy, and a reinforcement learning agent.

Technical Skills

Languages: Python, C/C++, CUDA, Go, Java, Verilog/SystemVerilog, MIPS Assembly
AI/ML Frameworks & Inference: PyTorch, vLLM, Transformers, torch profiler, Triton, CuTe DSL, LLVM
Tools & Profiling: Linux, Docker, NSYS, NCU, NVML, kineto traces
Core Competency Areas: Accelerated computing, deep learning, AI/ML optimization, model inference optimization, GPU kernels, HPC, computer architecture, compilers, in-memory computing, linear algebra, signal processing
Certifications: Fundamentals of Accelerated Computing in Modern CUDA C/C++ — Nvidia (July 2025), Advanced Machine Learning Bootcamp — Nittany AI Alliance (May 2025), Verilog HDL: VLSI Hardware Design — Udemy (July 2025)